

COMP8420 2026 S1

Main Project: Problem-based Group Project

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Assignment Details

Assignment Type: Problem-Based Group Project (4 students per group)

Weight: 40% of Final Grade

Content Coverage: Weeks 1–12

Release Date: Friday, 24 April 2026 (Week 7. The introduction, grouping and the use cases will be discussed in Week 7)

Presentation Due: Friday, 5 June 2026 (Week 13, at workshops, 1-7pm)

Report, Code & Video Due: Friday, 19 June 2026 (Exam Period)

Time Commitment: 30-50 hours per student

Submission Format: Four parts comprising:

- + Project report (PDF)
- + Project presentation (Powerpoint)
- + Project implementation (source codes, including code notebooks, repositories, etc.)
- + Project video (maximally 5-minute recording demonstrating the full system)

Learning Outcomes:

This assignment assesses the following unit learning outcomes:

- ULO1: Compare and evaluate the key natural language processing applications that meet the current and emerging industry needs.
- ULO2: Explain the main techniques that are used to develop and implement natural language processing applications.
- ULO3: Implement natural language processing applications using common tools and libraries used in industry.
- ULO4: Design natural language processing applications using advanced deep learning techniques.
- ULO5: Apply natural language processing methods and techniques to industry applications using real data.
- ULO6: Apply good practice in the development, monitoring, and deployment of natural language processing systems.

General Message:

It is expected that each student will spend approximately 30 to 50 hours to complete the project in a group. Each group consists of at least 3 students, suggested 4 students.

All members in the group should have significant and equal contributions to the project design, implementation, evaluation, and output, as well as to the presentation, and the report.

This assignment accounts for 40% of the overall assessment for this unit.

The unit lectures and workshops provide the NLP/LLM techniques, examples and strategies for you to undertake this problem-based real-life project. Start this assignment as soon as possible. You would be expected to plan and make weekly progress and discuss them with our tutors in the workshops if required.

Grouping:

In Week 7 (Friday, 24 April 2026), all students are requested to form your own groups (minimally 3, maximally 5 people in each group). Your task selection should be made by the Practice Workshop (by Friday, 1 May 2026) in Week 8.

If you cannot attend the workshop in that week, please send an email to inform the Teaching Associate (tutor) and the convener about your team members and project title by Week 9 (by Friday, 8 May 2026) at the latest.

1 Introduction

In this Main Project (a.k.a. Assignment 3), you work with your team members by taking different roles for the development and implementation of a problem-based Natural Language Processing (NLP) and large language model (LLM) project. You aim for generating NLP solutions using advanced NLP and LLMs for the real-world problem-solving.

Each group must select one Use Case from the six options below and develop an NLP/LLM system that must implement, integrate, and justify the NLP and LLM techniques to solve the problem. Your mission is to leverage the technologies and methodologies covered in this course, possibly in conjunction with additional tools, to craft a viable solution to the identified issue.

You are required to apply, compare, and justify your choice of techniques to solve a real-world problem with justified system design and solution (architecture, technology), problem-solving process and workflow, data (input and output), results, analysis, and recommendation. You may use your own preferred relevant datasets, or use one of the suggested datasets for each task below.

Your deliverables will include a detailed project report elucidating the problem-solving process, the problem-solving source codes, a team presentation in the class, and a video (maximally 5-minute recording) showing the full project demonstration from input to output.

2 Use Cases and Task Descriptions

Each group should just choose one of the following use cases, so please provide your preferences after we release them in Week 7 and we will discuss to consolidate your final use cases.

2.1 Use Case 1: Intelligent Customer Service Systems

- **Goal/tasks:** Build a customer service system that automatically categorizes inquiries, analyzes sentiment, extracts key information (such as customer/product names, issues), and uses such information to generate appropriate responses using NLP/LLMs.
- **Suggested Dataset:** [Microsoft Customer Service Dataset](#)
- **Basic Techniques:** Including text preprocessing, word embedding, NER for customer/product extraction, sentiment analysis with traditional ML, text classification for query categorization, POS tagging for understanding query structure, and TF-IDF for knowledge base matching.
- **Advanced Techniques:** Including prompting techniques (including prompting, chain-of-thought, ReAct, instruction, preference), agentic design, LLM (with foundations models) for response generation, unimodal to multimodal data, LLM prompt engineering techniques, RAG for company knowledge retrieval (if applicable), agentic modeling, dialogue state tracking, and automated escalation using LLM classification.

2.2 Use Case 2: Social Media Intelligent Platform

- **Goal/tasks:** Develop a system that monitors social media content, identifies trends, analyzes public sentiments, detects emerging topics, and generates insights for brand monitoring and campaign, etc.
- **Suggested Datasets:** [Sentiment140 Twitter Dataset](#) or [Reddit API](#) for real-time data
- **Basic Techniques:** Including text preprocessing, word embedding, NER for entities/locations, hashtag/mention extraction with regex, sentiment classification, trend analysis, influencer identification, and text clustering for detecting topics.
- **Advanced Techniques:** Including prompting techniques (including prompting, chain-of-thought, ReAct, instruction, preference), agentic design, RAG-based real-time content acquisition and moderation with LLMs (with foundation models) and trend explanation generation, crisis detection, multi-lingual analysis, branding strategy recommendation, and automated report generation.

2.3 Use Case 3: Research Paper Analysis and Recommendation System

- **Goal/tasks:** Create a system that analyzes academic papers, extracts metadata, summarizes findings, identifies research gaps, and recommends relevant papers.
- **Suggested Dataset:** [arXiv Dataset \(Kaggle\)](#)
- **Basic Techniques:** Including text preprocessing, word embedding, NER for authors/institutions, keyword extraction using TF-IDF, text classification for research domains, analyse mutual citations, and abstractive summarization with extractive methods.
- **Advanced Techniques:** Including prompting techniques (including prompting, chain-of-thought, ReAct, instruction, preference), RAG for paper search and related work discovery, LLM-based (with foundation models) paper summarization, research question summarization or generation, preference-based prompting, agentic design, LLM for multi-document synthesis, citation recommendation, and automated peer review assistance.

2.4 Use Case 4: Healthcare Text Analysis System

- **Goal/tasks:** Build a system that processes medical texts, extracts symptoms, treatments, and outcomes, and suggests recommendation, while ensuring privacy.
- **Suggested Dataset:** [Medical Transcriptions \(Kaggle\)](#) or [Drug Reviews Dataset](#)

- **Basic Techniques:** Including text preprocessing, word embedding, medical NER for symptoms/drugs/conditions, sentiment analysis for patient satisfaction, text classification for medical specialties, and relation extraction for symptom-treatment pairs.
- **Advanced Techniques:** Including prompting techniques (including prompting, chain-of-thought, ReAct, instruction, preference), agentic design, text classification, medical question answering with prompting and RAG, LLM generation for clinical suggestion and recommendations, privacy-preserving text analysis, and medical literature summarization.

2.5 Use Case 5: Legal Case Citation Analysis System

- **Goal/tasks:** Develop a system that analyzes legal case citations, classifies citation relationships (cited, referred to, applied, followed), predicts case outcomes, and provides legal precedent analysis.
- **Suggested Dataset:** [Legal Citation Text Classification Dataset](#)
- **Basic Techniques:** Including text preprocessing, word embedding, legal NER for case names/citations/dates, text classification for citation types (cited, referred to, applied, followed), case outcome prediction using traditional ML, key phrase extraction from legal reasoning, and TF-IDF similarity for finding related cases.
- **Advanced Techniques:** Including prompting techniques (including prompting, chain-of-thought, ReAct, instruction, preference), RAG for retrieval, agentic design, LLM-based legal case reasoning summarization, citation relationship explanation, legal precedent recommendation, automated case outcome prediction with reasoning, and legal argument generation.

2.6 Use Case 6: E-commerce Intelligent System

- **Goal/tasks:** Create a system to extract product aspects and sentiments, analyze product reviews, detect fake reviews, generate product descriptions, and provide business insights.
- **Suggested Dataset:** [Amazon Product Reviews](#)
- **Basic Techniques:** Including text preprocessing, word embedding, sentiment analysis, product NER, review classification, fake review detection using traditional ML, and price/feature extraction.
- **Advanced Techniques:** Including prompting techniques (including prompting, chain-of-thought, ReAct, instruction, preference), RAG retrieval, LLM-based summarization of customer/product reviews, product description summarization or generation, review summarization or classification, competitor analysis, dynamic pricing insights, agentic design, and personalized recommendations and product popularity analysis.

3 Technical Requirements

All techniques taught in the lessons are relevant and could be considered in your implementation. You are encouraged to demonstrate appropriate but at least **THREE** Basic and Three Advanced NLP techniques (for basic marks), from those given below or others. The choices may vary and depend on your use case scenarios, which should be justified (compared, and analyzed against the use case).

3.1 Basic Techniques

- Text Preprocessing Pipeline (tokenization, normalization, stemming/lemmatization)
- Named Entity Recognition (spaCy, NLTK, or custom)
- Part-of-Speech Tagging (syntactic analysis)
- Traditional Text Classification (Naive Bayes, Logistic Regression with BoW/TF-IDF)
- Sentiment Analysis (traditional ML)
- Text Similarity and Clustering (Embedding cosine similarity, Sentence Embeddings)
- Rule-based Information Extraction (pattern matching, regex)

3.2 Advanced Techniques

- LLM (foundation) models (GPT, Gemma, Llama, etc.)
- Retrieval Augmented Generation (RAG)
- Prompting engineering (templates, ReAct, instructions, or feedback, etc)
- Chain-of-Thought prompting
- Automated evaluation (LLM-as-a-judge)
- Fine-tuning/Adaptation techniques (Few-shot Learning, etc)
- Ensemble Methods (combining models)
- Agentic design
- Multimodal LLM
- LLM with privacy, fairness or ethics considerations

4 Project Submission and Presentation Guidelines

4.1 Submission Requirements

We anticipate that each group will fulfill four primary obligations per the following detailed Submission Requirements:

1. **Group Presentation:** A brief presentation with powerpoint is expected to be delivered by all group members at the Practice Workshop in Week 13 (1-7pm, Friday, 5 June 2026). It is imperative that all group members actively demonstrate your contributions in the project through the presentation. Your group presentation spans 3-4 minutes, maximally 5 minutes, otherwise with overtime mark reduction. Under exceptional cases without other solutions, proposals to present in Week 12 may be allowed per justification and approval. Group presentation **CANNOT** be postponed after Week 13.
 - The group presentation may briefly outline the target problem, objectives, methodology, basic and advanced techniques, data, results and analysis, findings and discussion.
 - Each group must complete your project presentation within 5 minutes (maximally 5 minutes, mark reduction would be incurred for overtime), at the Week 13 Practice Workshop.
 - The presentation file (powerpoint) should be submitted to iLearn before Friday, 5 June 2026.
 - Each group member must join the group presentation, and should present and demonstrate your specific contributions to the project.
 - An outline of key roles and responsibilities of each group member should be included in the presentation.
2. **Project report:** Should be submitted to iLearn.
 - Basic report structure: Introduction, Problem and Task, Roles and Responsibilities, Methodology, Data, Experiment and analysis, Recommendation and discussion, Conclusion, Reference, etc.
 - Project title and group information on the first page.
 - Introduction: what background, real-world challenges, and NLP/LLM techniques will be addressed in your project, and explain their business and technical significance.
 - Problem and Task: introduce the use case, its challenges to be addressed, related work and gap analysis, your task decomposition in this project.
 - Roles and Responsibilities: group members and their details (name, student ID, contact emails, etc.), roles and responsibilities and main contributions of each member in this project.

- Methodology: foundation models, system architecture, system workflow, basic and advanced NLP/LLM technologies chosen and implemented, connectivity between processes and techniques, etc. Justifications of chosen model, architecture, workflow, and techniques should be made in the context of the use case challenges and pros and cons of each technique and other alternatives that you do not choose.
 - Data and evaluation: datasets utilized, explain their structures, characteristics, complexities, text preprocessing, and data management systems; baseline models, evaluation measures for performance, etc. Explain why you choose the data, baselines, and evaluation in the context of other alternatives.
 - Experiment and analysis: experimental results (tables, diagrams), demonstrate the design effectiveness, efficiency, etc; empirical or analytical analysis of the results in comparison to alternative solutions; analyze the results against your methodology design.
 - Recommendation and discussion: what insights and findings are identified through this project, what are the limitations of the methodology, data, implementation, etc., what will be considered in future work, etc.
 - Resources: provide resource links of your project (Github etc), including your models, source codes, results, and video. Make sure your markers can easily identify them and download your project and models.
3. **Source codes and datasets:** Should be accessible to the markers (e.g., maintaining a public Github repo and submitting an URL for docker images).
- Source codes: Structure your project as a series of small programs or Jupyter notebooks. Include a README file to guide users through your project.
 - Datasets: Document the source of your dataset (for reused datasets) or the collection process (for original datasets). Submit a portion of the dataset (less than 5MB if possible) for verification.
 - Models: If your models are too large for direct submission, provide a download link within your report.
4. **Video:** Should be accessible to the markers (e.g., providing a public Github repo and submitting an URL).
- Recording: The video should demonstrate the entire workflow of the system, from input to output. The demo may highlight key NLP techniques, insights, contributions, analysis, justification, and findings.
 - All group members would be expected to contribute to the recording.
 - Format: use generally accessible formats.
 - Storage: If video is too big for direct submission, provide a download link in your report.

4.2 Deliverable Formats

Your deliverables from this group project should comprise four components as follows: Project Report, Group Presentation, Project Implementation, and Project Video.

1. Project Report

```
GroupID_Assignment3/  
    ...  
    Report/  
        GroupID_Report.pdf
```

- Maximum 5,000 words in the PDF format, placed under **Report/** as **GroupID_Report.pdf** as shown above.
- Submissions that do not follow this structure or without the report will incur penalty (up to 13 marks).

2. Project Presentation

```
GroupID_Assignment3/  
    ...  
    Presentation/  
        Presentation.pdf
```

- Include your powerpoint presentation you use in the class presentation as shown above.
- Submissions that do not follow this structure or without the powerpoint presentation will incur penalty (up to 7 marks).

3. **Project Implementation** Show your implemented system with workflow demonstrations and explanation. Typically, the deliverables may be presented in Jupyter Notebooks to demonstrate your implementation. User interfaces, data management systems, output visualization would be expected to accompany the demonstrations.

```
GroupID_Assignment3/  
    Codes/  
        Notebook1.ipynb  
        Notebook2.ipynb  
        Result files  
    ...
```

- Submissions that do not follow this structure may incur penalty (up to 5 marks).

- No to limited implemented systems and results will incur penalty (up to 20 marks).
4. **Project Video** A recording shows your implemented system with workflow demonstrations and explanation, system development environment, user interfaces, data management, and output visualization, etc.

```

GroupID_Assignment3/
  Video/
    Video files
    ...

```

- Submissions that do not follow this structure or without vide may incur penalty (up to 2 marks).

4.3 Other Issues

- **Submission Method:** Upload a single zip file containing the project report, presentation, implementation, and video to iLearn.
- **Academic Integrity:** All codes must be your original work with proper citations for external resources if applied. Using generative AI tools without your own implementation for the project will incur heavy penalty.
- **Reproducibility:** Ensure all results can be reproduced.

Note that:

- You should **NOT** recycle the assignments or any related materials completed in past or concurrent ongoing courses.
- Focus on industry scenarios and an applicable system would be appreciated.
- Any actions that violate MQ's policy academic integrity policies are strictly prohibited.
- Please set your GitHub repository (if you use it) to private before the submission deadline (to avoid plagiarism), and remember to make it public immediately after the deadline.

5 Grading Rubric (Total: 40 marks)

The total score of this group project comprises three parts: project implementation (with system demonstrations and video, up to 20 marks), project report (up to 13 marks), and project presentation (up to 7 marks), as explained below.

In these deliverables, you are expected to demonstrate how your system has sufficiently addressed the following key aspects.

5.1 Project Implementation (up to 22 points)

Your source codes and working system should demonstrate what basic and advanced NLP/LLM techniques have been developed, analyzed and justified to address the selected use case problem.

5.1.1 Basic Techniques (up to 6 marks)

Component	Marks	Criteria
Justified selection of NLP techniques	1	Justified selection of 3 basic techniques (0.5); More techniques analyzed and justified (0.5)
Implementation and integration	3	Correct implementation of basic techniques (2), demonstrated successful integration into the system with analysis and justification (1)
Evaluation and analysis	2	Complete experimental results with quantitative evaluation (1), comparison and analysis of results against alternative techniques with discussion on limitation and improvement against the use case complexities (1)

5.1.2 Advanced Techniques (up to 11 marks)

Component	Marks	Criteria
LLM foundation models	2	Redeveloped foundation models (2), finetuned LLM foundation models (1.5), or adoption and integration of pretrained models (0.5)
Justified selection of LLM processing techniques	2	Justified selection of 5 LLM techniques (1), and more LLM techniques analyzed and justified (1)
Implementation and integration	4	Sufficient and effective implementation of LLM techniques (3), and demonstrated successful system integration with those basic techniques (1)
Evaluation and analysis	3	Complete experimental results with quantitative evaluation (1), comparison and analysis of results with discussion on limitation and improvement against the use case complexities (1), comparison analysis with other foundation and NLP models and techniques (1)

5.1.3 System Integration and Performance (up to 5 marks)

Component	Marks	Criteria
User interfaces	1	User-friendly user interfaces for input, output, and interactions
Cohesive design	1	All basic and advanced techniques are effectively and smoothly integrated into underlying foundation models
System performance	1	Efficient and effective operations
Operations	1	Smooth operations of the entire workflow and process
User experience	1	Very positive user experience in using the system

5.2 Project Report (up to 11 points)

The report should demonstrate basic and advanced NLP/LLM techniques implemented in the project and address the following aspects sufficiently and professionally.

Methodology and System (up to 3 marks)

Component	Marks	Criteria
System architecture and workflow	1	Clear and logical pipeline integrating basic and advanced techniques, clear explanation of the system workflow and process
Data selection and preparation	1	Justified dataset selection for the use case (0.5), data preprocessing (0.5)
Comparative analysis	1	Comparison of different techniques (0.5), justification for final choices (0.5)

Experiment and Evaluation (up to 5 marks)

Component	Marks	Criteria
Results presentation	2	Clear multifaceted structured (1) and visual (1) presentations of results from different aspects
Evaluation and analysis	2	Quantitative and qualitative evaluation against baseline methods (1.5), and justification addressing the use case complexities (0.5)
Critical discussion and reflection	1	Insightful discussion about findings (0.5), and limitation and improvement (0.5)

Organization and Presentation (up to 3 marks)

Component	Marks	Criteria
Report organization	1	A well organized report following the suggested report structure, with professional report design
Presentation	2	Effective and clear statement of the report (1), and convincing evaluation, analysis, evidenced justification, insightful discussion about findings, limitation and improvement (1)

5.3 Presentation (up to 5 marks)

Component	Marks	Criteria
Content Coverage	2	Clear organization and explanation of the problem, methodology (system architecture, workflow), NLP techniques, data, implementation, evaluation, experiment and results
Demonstration	1	Show a working system, workflow, and functions of NLP techniques
Delivery	0.5	Engaging, well-paced, clear communication
Teamwork	0.5	All members contribute, good team collaboration in the implementation, report, and presentation
Individual contributions	0.5	Individual contribution to the project implementation, report, and presentation
Q & A	0.5	Performance in answering questions and defending choices

5.4 Video (up to 2 marks)

Component	Marks	Criteria
Function demonstration	1	Effective demonstration of system implementation, functions and nonfunctional performance
Interpretation	1	All team members contribute to the effective introduction to the system

Notes:

- Presentation will be held in Week 13. All group members must participate in the presentation.
- Higher marks correspond to 1) more advanced techniques, 2) more justified analysis and evaluation, 3) effective team collaboration, 4) professional design, implementation, report and presentation, and 5) deep analysis and insights.
- Submissions that do not follow the submission and presentation guidelines will incur penalty (up to 5 marks).
- Both report and presentation contribute significantly to the overall assessment. For the report, marks are awarded to both implementation and analysis.
- Students can reuse the tools and examples covered in the lab exercises with substantial redevelopment.

- Students are refrained from the direct use of LLM platforms such as ChatGPT and Gemini for code generation.
- Plagiarism checks will be made, and those which appear to have academic integrity issues will be reported. [Academic Integrity Policy \(mq.edu.au\)](https://mq.edu.au/academic-integrity-policy).

6 Policies and Notes

6.1 Late Submission Policy

For both assignments and the major project, NO late submission policy applies, unless Special Consideration has been submitted and approved. All deadlines mean 11:59:59 pm that day.

- **5% penalty per day:** If you submit your assessment late, 5% of the total possible marks will be deducted for each day (including weekends), up to 7 days.
 - Example 1 (out of 100): If you score 85/100 but submit 20 hours late, you will lose 5 marks and receive 80/100.
 - Example 2 (out of 30): If you score 27/30 but submit 1 day late, you will lose 1.5 marks and receive 25.5/30.
- **After 7 days:** Submissions more than 7 days late will receive a mark of 0.
- **Extensions:**
 - Automatic short extension: Some assessments are eligible for automatic short extension. If so, you can only apply for an automatic short extension before the due date.
 - Special Consideration: If you need more time due to serious issues and for any assessments that are not eligible for Short Extension, you must apply for Special Consideration.

6.2 Special Consideration Policy

The Special Consideration Policy aims to support students who have been impacted by short-term circumstances or events that are serious, unavoidable and significantly disruptive, and which may affect their performance in assessment. If you experience circumstances or events that affect your ability to complete the assessments in this unit on time, please inform the convenor and submit a Special Consideration request through ask.mq.edu.au.